

CurbRisk-AV — Quantifying autonomous vehicle pickup and drop-off risk using exposure-adjusted complaint data

Final Paper

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Paper Under Construction

Planned completion by ...

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Abstract

TODO: abstract goes here (write last).

1 Introduction

1.1 Problem Statement

1.1.1 The Problem:

Autonomous-vehicle pick-up / drop-off (PUDO) maneuvers — the moments a driverless vehicle stops so a rider can get in or out — are a plausible source of curb-space conflict, minor collisions, and rider complaints. Whether that risk is growing, shrinking, or holding steady as a fleet scales is hard to answer from the public record: California Public Utilities Commission (CPUC) AV Deployment reports are large, redacted, and change format from quarter to quarter, so nobody has simply lined them up into a rate over time.

1.1.2 Contribution

- A reproducible pipeline (**pudo-pipeline**) that ingests nine CPUC quarterly report zips for Waymo’s driverless deployment — two incompatible file schemas, redaction, and formatting drift included — into five analysis-ready tables.
- An exposure-adjusted PUDO complaint rate across eight quarters (2024 Q3 – 2026 Q2), on trips — the team’s chosen exposure measure, cross-checked against total vehicle miles traveled — with an explicit trend test (Poisson / Negative-Binomial GLM), a distribution-free cross-check (Mann-Kendall), and a Bayesian refit (PyMC/NUTS) that replaces the frequentist p-value with a direct posterior probability the rate is declining.
- A placement of the PUDO hazard on a probability $\times$ impact risk matrix that is computed from the data rather than asserted, with the probability basis left as a visible, toggleable assumption.

1.1.3 Scope and Vision

Scope: one carrier (Waymo), one state (California), eight quarters, public CPUC filings only — no proprietary telemetry. **Vision:** the pipeline’s loaders are written against the CPUC’s public template, not against Waymo specifically, so the same numerator/denominator/trend method can be pointed at any other CPUC-reporting AV carrier to support carrier-to-carrier comparison rather than a single-carrier trend.

1.2 Technical Challenges

- **Two incompatible numerator schemas.** 2024 reports ship one aggregate, pre-summed row per quarter; 2025Q1 onward ship ride-level microdata (millions of rows) with a Y/N complaint flag. Both had to be reconciled into one numerator, and the level shift between eras cannot be fully ruled out.

- **Redaction.** Incident timestamps, locations, and the `PUDOTravelLane` flag are blank in every file of every period, ruling out within-quarter timing or spatial analysis.
- **Silent formatting drift.** Comma-grouped numerics, filenames that change across quarters, and a transposed carrier ID silently nulled the exposure data until caught — the loader now warns loudly instead of nulling quietly.
- **Non-calendar reporting periods.** The 2024 CPUC reports are not calendar quarters, so “2024 Q3” / “2024 Q4” needed explicit relabeling (`period_label`, `is_calendar_quarter`) to avoid silently comparing the wrong months.
- **Small, overdispersed counts.** 27–79 complaints per quarter is enough to rule out the normal-approximation trend test, but not enough to ignore quarter-to-quarter noise.

2 Related Work

3 Data

3.1 Data Sources

Public data comes from the CPUC AV Program’s quarterly deployment reports (cpuc.ca.gov → Licensing → Autonomous Vehicle Programs → Quarterly Reporting) [1]. Two file families, both filed per reporting period:

- **Incidents-Complaints** — the numerator. PUDO complaint and PUDO-collision counts.
- **Month-Level** — the denominator. Real `Year/Month`, trips, and vehicle miles traveled (VMT). VMT is further split into three “period” columns in the source file, but CPUC’s own reference key never defines what those periods mean, so this analysis uses trips as the primary exposure measure and VMT only as an unsplit, whole-quarter cross-check.

Nine raw reporting windows (Jun–Aug 2024, Sep–Nov 2024, a Dec 2024 one-month stub, then eight calendar quarters through 2026 Q2) are recombined into eight analysis quarters, 2024Q3–2026Q2.

The pipeline’s two upstream file families reduce to a single hand-off table, `final_pudo_summary` — one row per analysis quarter, carrying only the complaint numerator and the trips denominator (Table 1). This is the dataset the rest of the paper’s analysis runs on:

Table 1: The final hand-off dataset (`final_pudo_summary`): PUDO complaint counts and trip exposure by analysis quarter, 2024 Q3 - 2026 Q2 ($n = 8$).

period	trips	PUDO complaints
2024 Q3 (Jun-Aug)	751,844	41
2024 Q4 (Sep-Dec)	1,852,614	51
2025 Q1	1,818,206	27
2025 Q2	2,234,912	42
2025 Q3	2,676,001	42

period	trips	PUDO complaints
2025 Q4	3,748,532	66
2026 Q1	3,905,934	69
2026 Q2	4,220,075	79

All numbers, tables, and figures in this paper are pulled directly from the project’s analysis notebook — see [pudo_analysis.ipynb](#) for the full code and every intermediate result.

4 Methods

4.1 Feature Engineering

Every analysis period carries a clean sort/filter key (`period_id`) alongside a human-readable label, its exact date window, and a flag for whether that window is actually a calendar quarter (`period_label`, `window_start/window_end`, `is_calendar_quarter` — false only for the two 2024 periods). Two exposure denominators are computed per period — trips (one pick-up + one drop-off per trip, the most literal PUDO-opportunity count; the primary measure) and total VMT (a cross-check) — each paired with a `pudo_per_100k_*` rate, plus a PUDO-share-of-all-complaints column for context.

4.2 Modeling

4.2.1 Exposure-adjusted count model

The complaint count Y_t in analysis quarter $t = 0, \dots, 7$ is modeled as a Poisson process whose rate scales with exposure E_t (trips, or VMT as a cross-check), with a log-linear time trend:

$$Y_t \mid \mu_t \sim \text{Poisson}(\mu_t), \quad \log \mu_t = \log E_t + \beta_0 + \beta_1 t \quad (1)$$

$\log E_t$ enters with a fixed coefficient of 1 (an *offset*), so β_1 is the log change in the underlying **rate** μ_t/E_t per quarter, not in the raw count. Every “per-quarter % change” quoted in this paper is:

$$\% \text{ change per quarter} = (e^{\beta_1} - 1) \times 100\% \quad (2)$$

4.2.2 Overdispersion check

Equation 1’s defining assumption is $\text{Var}(Y_t) = \mathbb{E}[Y_t]$. This is checked with the index of dispersion (variance-to-mean ratio, VMR) on the raw complaint counts across the $n = 8$ quarters:

$$\text{VMR} = \frac{s^2}{\bar{y}}, \quad D = (n - 1) \text{VMR} \sim \chi_{n-1}^2 \quad \text{under } H_0 : \text{Poisson} \quad (3)$$

VMR ≈ 1 is consistent with a pure Poisson process; VMR $\gg 1$ — the result found here — rejects equidispersion in favor of a Negative-Binomial model.

4.2.3 Negative-Binomial GLM

When Equation 3 rejects equidispersion, Y_t is instead modeled as Negative-Binomial (the NB2 parametrization used by `statsmodels`), keeping the same log-linear mean structure as Equation 1:

$$Y_t \mid \mu_t, \alpha \sim \text{NB}(\mu_t, \alpha), \quad \text{Var}(Y_t) = \mu_t + \alpha \mu_t^2, \quad \log \mu_t = \log E_t + \beta_0 + \beta_1 t \quad (4)$$

with the dispersion parameter α estimated by maximum likelihood alongside β_0, β_1 . Larger α means *more* overdispersion relative to Poisson under this parametrization.

4.2.4 Exact (Garwood) Poisson confidence interval

Per-quarter rate uncertainty is reported with the exact Poisson interval rather than the normal/Wald approximation, so it stays valid at small or zero counts:

$$\text{CI}_{1-\alpha}(k) = \left[\frac{1}{2} \chi_{\alpha/2}^2(2k), \frac{1}{2} \chi_{1-\alpha/2}^2(2k+2) \right] / E \times 10^5 \quad (5)$$

where $\chi_p^2(\nu)$ is the p -quantile of a chi-square distribution with ν degrees of freedom, k the observed complaint count, and E the exposure (trips) for that quarter.

4.2.5 Distribution-free trend check

As a check that assumes no particular count distribution, Kendall's τ is computed between the quarter index and the per-quarter rate $r_1, \dots, r_n$ (chronological order):

$$\tau = \frac{2}{n(n-1)} \sum_{i < j} \text{sign}(r_j - r_i) \quad (6)$$

$\tau < 0$ agrees with a declining trend; its p -value is reported alongside, separately from the GLM's.

4.2.6 Bayesian Negative-Binomial refit

With only eight quarters, Equation 4's confidence interval is built by estimating α by maximum likelihood and then treating it as *known* — a step that understates uncertainty in a small sample. As a check, the same mean structure is refit as a full posterior (PyMC, NUTS sampler) with weakly informative priors:

$$\begin{aligned}
\beta_0 &\sim \mathcal{N}(\log(\bar{r}), 2^2) \\
\beta_1 &\sim \mathcal{N}(0, 1^2) \\
\alpha &\sim \text{Exponential}(1) \\
\mu_t &= E_t \exp(\beta_0 + \beta_1 t) \\
Y_t \mid \mu_t, \alpha &\sim \text{NB}(\mu_t, \alpha), \quad \text{Var}(Y_t) = \mu_t + \mu_t^2/\alpha
\end{aligned} \tag{7}$$

where $\bar{r}$ is the pooled baseline rate and $\beta_1 \sim \mathcal{N}(0, 1)$ deliberately expresses no prior belief about direction. Note Equation 7’s variance term is the **inverse** sense of Equation 4’s α (larger α here means *closer* to Poisson) — the two are not numerically comparable. The model is sampled with 4 chains $\times$ 2000 draws, checked for convergence ($\hat{R}$, divergence count) and for sensitivity to the prior width on β_1 . The quantity of interest is the posterior probability the rate is actually declining, estimated directly from the fraction of posterior draws below zero rather than from a p -value:

$$P(\text{declining}) = P(\beta_1 < 0 \mid y_{1:n}) \tag{8}$$

4.2.7 Risk-matrix probability basis

To place the PUDO hazard on a probability $\times$ impact risk matrix, three ways of expressing “how likely is a PUDO event” are compared, holding the impact reading fixed. Let $k = \sum_t y_t$ be total complaints and $N = \sum_t E_t$ total trips over the window.

Per-period — the probability of at least one complaint in a reporting quarter, treating the mean quarterly count $\bar{\lambda} = k/n$ as a Poisson rate:

$$p_{\text{per-period}} = 1 - e^{-\bar{\lambda}} \tag{9}$$

Per-trip — the raw empirical per-trip probability:

$$p_{\text{per-trip}} = \frac{k}{N} \tag{10}$$

Bayesian — a Beta-Binomial posterior on the per-trip rate (Jeffreys-style Beta(0.5, 0.5) prior), converted to a predictive probability of at least one complaint in a future quarter of size E_{next} :

$$\begin{aligned}
\text{rate} \mid k, N &\sim \text{Beta}(0.5 + k, 0.5 + N - k) \\
p_{\text{bayesian}} &= 1 - e^{-\widehat{\text{rate}} \cdot E_{\text{next}}}
\end{aligned} \tag{11}$$

using the posterior mean $\widehat{\text{rate}}$ and the most recent quarter’s trip volume as E_{next} .

5 Evaluation and Results

Per-quarter descriptive statistics for the complaint counts, exposure, and rate (Table 2) show why raw counts cannot be compared directly: exposure grows nearly $6\times$ across the window.

Table 2: Per-quarter descriptive statistics for PUDO complaints, trips, and the PUDO-per-100k-trips rate, 2024 Q3 - 2026 Q2 ($n = 8$).

variable	mean	std	min	median	max	range	CV
PUDO complaints	52.125	17.57	27.0	46.5	79.0	52.0	0.337
trips (M)	2.651	1.216	0.752	2.455	4.22	3.468	0.459
rate /100k trips	2.317	1.324	1.485	1.819	5.453	3.968	0.572

The index-of-dispersion test (Equation 3) on these counts gives $\text{VMR} \approx 5$ ($D \approx 41.5$ against χ^2_7 , $p < 0.001$), rejecting Poisson equidispersion — the reason every trend fit below uses the Negative-Binomial model of Equation 4 rather than Equation 1 directly. The per-quarter rates in Table 3 and throughout use the exact interval of Equation 5, not a normal approximation.

Pooling across all eight quarters gives the headline exposure-adjusted rate (Table 3): Over the study window Waymo averaged **1.97** PUDO complaints per 100,000 trips (exact Poisson 95% CI 1.78-2.16), from 417 complaints over 21.2M trips.

Table 3: Pooled PUDO complaint rate over the study window (417 complaints, 2024 Q3 - 2026 Q2).

complaints	trips (M)	rate per 100k trips	95% CI
417	21.2	1.966	[1.782, 2.164]

PUDO complaints are also worth reading in context of everything else riders report. Figure 1 stacks the PUDO count against every other reported complaint category, per quarter:

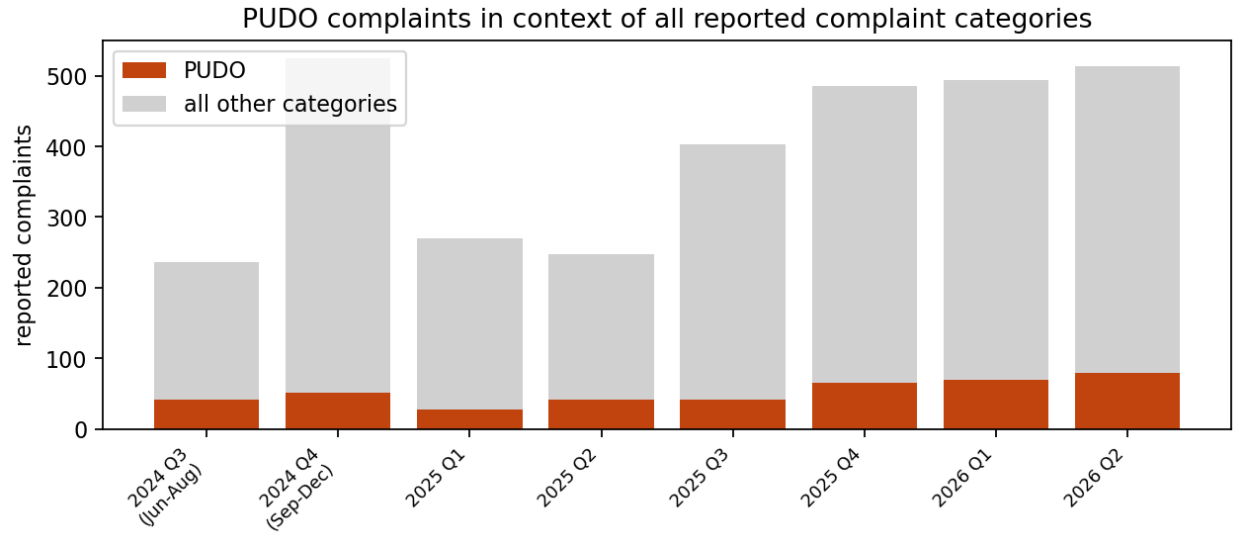


Figure 1: PUDO complaints against all other reported complaint categories, per quarter.

PUDO complaints average **13%** of all reported complaint categories over the window (range 10%-17%), making PUDO the largest location/behaviour complaint category (Safety and ‘Other’ are larger overall categories, but are not location-specific).

PUDO complaints scale with total complaint volume — the PUDO share stays in a fairly narrow band rather than drifting — so PUDO is not a shrinking or growing slice of the overall complaint mix; it is consistently the largest category tied to a specific vehicle maneuver rather than a general safety or “other” catch-all.

Fitting the Negative-Binomial GLM of Equation 4 under both exposure denominators — trips (primary) and VMT total (cross-check) — and reading the slope off Equation 2 gives the same direction (Table 4), with Kendall’s τ (Equation 6) agreeing: The exposure-adjusted trend is negative under all 2 exposure denominators tested (per-quarter change -13% to -11%; GLM p 0.001-0.007), and Mann-Kendall’s tau is negative under all of them as well, though only marginally significant (p 0.40-0.55) given the non-monotone, fell-then-flat shape of the series.

In more detail: Under the primary trips denominator, the fitted Neg. binomial slope is -10.7% per quarter (95% CI -17.8% to -3.1%, p = 0.007; total change -55% across the window).

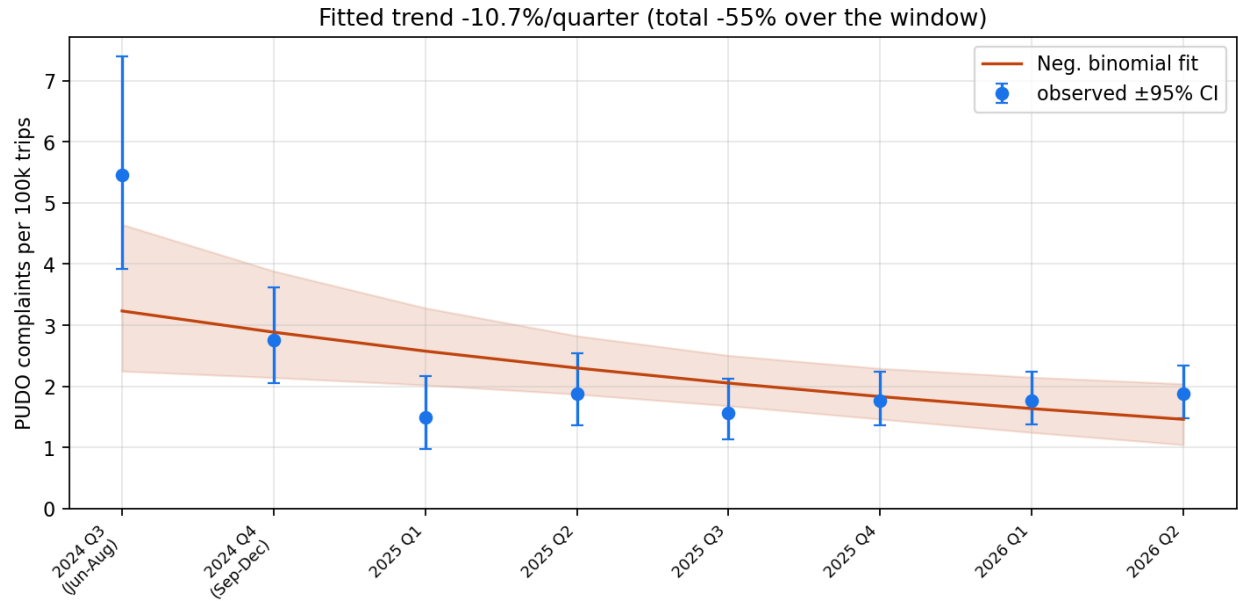


Figure 2: Observed PUDO complaint rate (per 100k trips, $\pm 95\%$ exact Poisson CI) against the fitted trend line and its 95% confidence ribbon, under the primary trips denominator.

Figure 2 shows what that decline actually looks like: a sharp drop from the 2024 Q3 launch-ramp quarter, then a rate that wanders inside a fairly narrow, overlapping band of confidence intervals for the rest of the window — a fit that is falling on average, but front-loaded rather than a steady quarter-over-quarter slide.

Table 4: Poisson / Negative-Binomial trend fit and Mann-Kendall check under every exposure denominator.

denominator	model	%/qtr	95% CI	p (GLM)	total	MK tau	p (MK)
Trips	Neg. binomial	-10.7%	[-18%, -3%]	0.007	-55%	-0.21	0.548
VMT total	Neg. binomial	-12.7%	[-20%, -5%]	0.001	-61%	-0.29	0.399

That decline is almost entirely the 2024 launch ramp — dropping that quarter cuts the slope by roughly two-thirds and the effect is no longer statistically significant (see Limitations). From 2025 Q1 onward the rate is essentially flat.

The Bayesian refit (Equation 7, Table 5) tells a consistent but more cautious story, reporting Equation 8 directly rather than a p -value: A Bayesian Negative-Binomial fit (weakly informative priors, PyMC/NUTS, 4 chains x 2000 draws, 0 divergences, max R-hat 1.00) gives a posterior median slope of **-10.9%** per quarter (95% credible interval [-25.2%, +7.6%]) and **P(the rate is declining) = 90%** — consistent with, but appropriately less certain than, the frequentist $p = 0.007$.

Table 5: Bayesian Negative-Binomial trend model posterior summary (PyMC/NUTS, 4 chains x 2000 draws, 0 divergences).

parameter	mean	sd	hdi_2.5%	hdi_97.5%	r_hat	ess_bulk
b0	-10.282	0.394	-11.059	-9.529	1.0	2731.0
b1	-0.115	0.091	-0.299	0.058	1.0	2741.0
alpha	3.131	1.502	0.649	6.036	1.0	2921.0

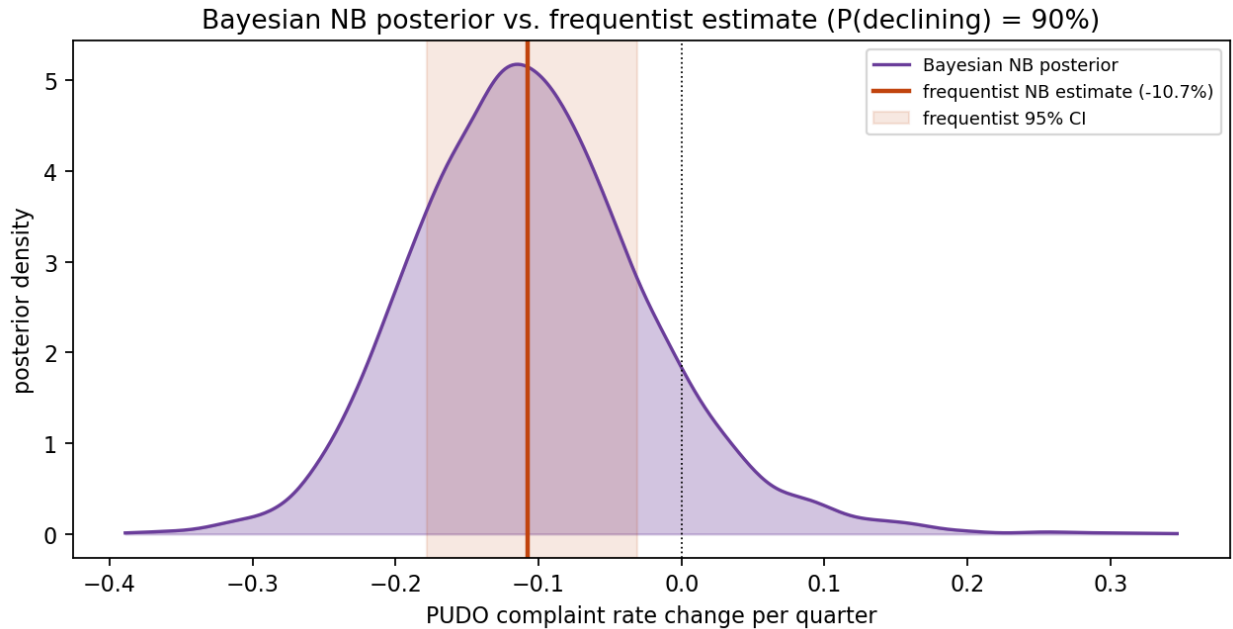


Figure 3: Bayesian Negative-Binomial posterior on the per-quarter rate change, vs. the frequentist point estimate and 95% CI.

The posterior barely shifts across a $4\times$ range of prior width on the slope (Table 6) — the conclusion is driven by the data, not the choice of prior:

Table 6: Bayesian NB posterior sensitivity to the prior width on the trend slope.

prior sd on slope	posterior median %/qtr	P(declining)
0.5	-11.0%	90.0%
1.0	-10.8%	90.6%
2.0	-11.1%	90.6%

To place the hazard on a risk matrix, the consequence of a PUDO event is read off a risk curve — how the event rate falls as severity rises (Figure 4). Nothing severe or fatal was observed at the PUDO level over the window.

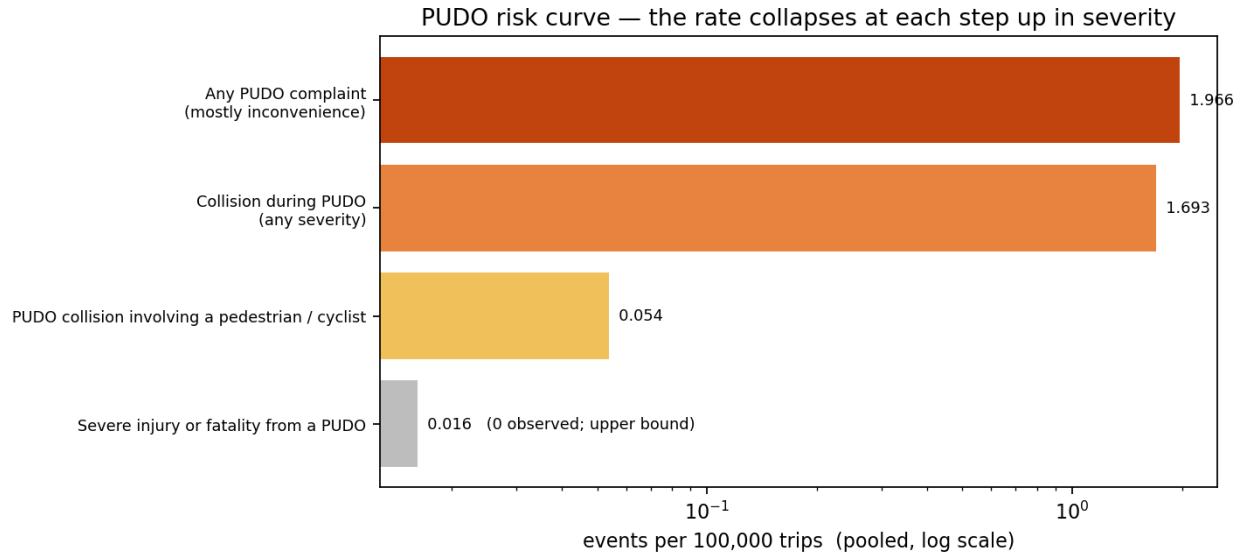


Figure 4: PUDO risk curve: event rate per 100,000 trips at each step up in consequence severity (log scale). The unfilled bar is a rule-of-three upper bound on a rate with zero observed events, not a measurement.

Combining that impact reading with a probability of a PUDO complaint occurring in a given reporting quarter — Table 7 compares the three formulas of Equation 9, Equation 10, and Equation 11 — places the hazard on the matrix as follows: Probability: **Almost certain** ($P(\geq 1 \text{ PUDO complaint in a quarter}) = 1.0000$). Impact: **Minor** (the worst consequence tier clearing 0.1/100k trips in the risk curve above). Combined risk level: **Medium**.

Table 7: PUDO risk-matrix placement under three probability bases (impact held fixed).

prob basis	probability	impact	risk level
per_period	Almost certain	Minor	Medium
per_trip	Moderate	Minor	Low
bayesian	Almost certain	Minor	Medium

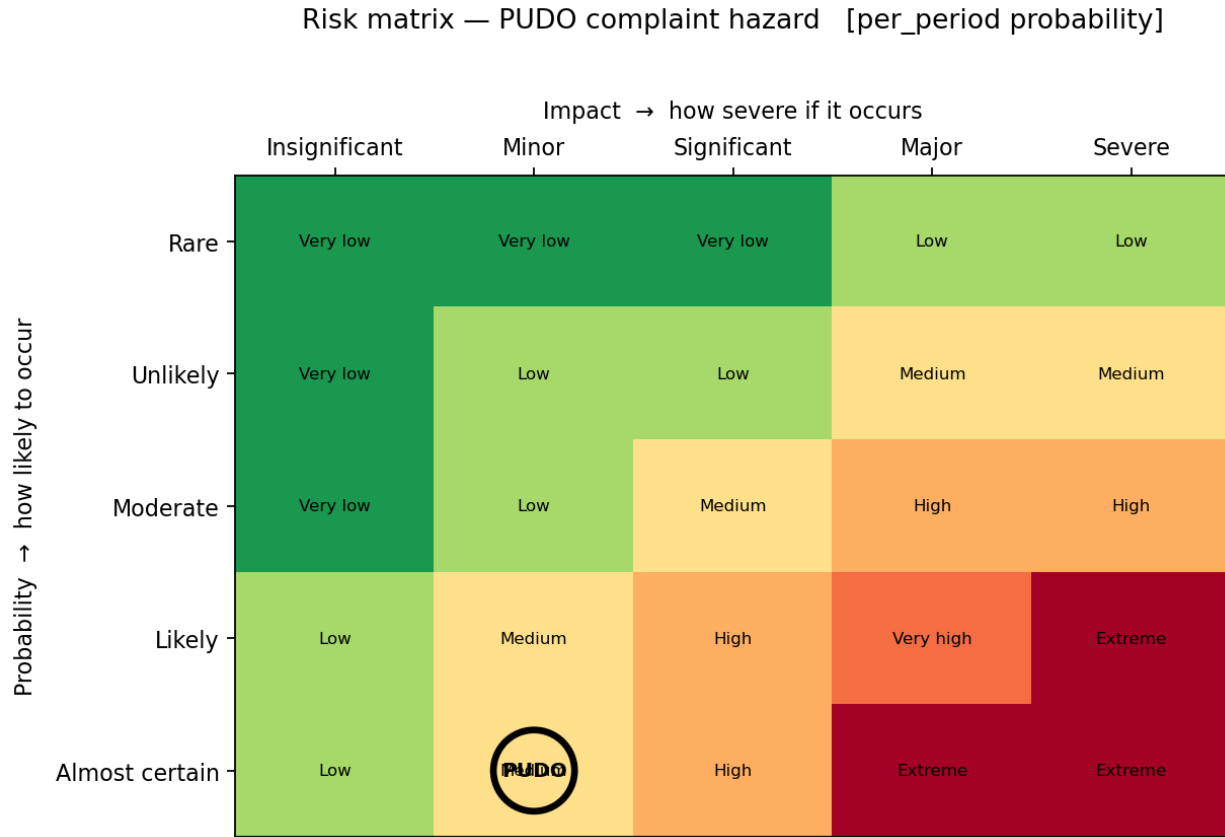


Figure 5: PUDO complaint hazard on a probability $\times$ impact risk matrix, placed from the data under the `per_period` probability basis.

6 Discussion

6.1 Key Concepts and Definitions

- **PUDO** — pick-up / drop-off: the maneuver where a vehicle stops to let a rider board or alight.
- **Exposure denominator** — the activity measure a complaint count is divided by (trips or VMT) so rates are comparable across quarters of very different fleet size.
- **Overdispersion** — when a count’s variance exceeds its mean beyond what a Poisson process allows, indicating real between-period rate variation.
- **Risk matrix axes** — *Probability*: how likely the hazard is to occur in a given period; *Impact*: how severe the outcome is when it does.
- **Credible interval vs. confidence interval** — the Bayesian model’s 95% credible interval is a direct probability statement (“there is a 95% chance the true slope is in this range, given the data and the prior”); the frequentist confidence interval is not, which is why the two are reported side by side rather than treated as interchangeable.

6.2 Assumptions

- PUDO complaints are treated as a Poisson-type count process for modeling purposes; the overdispersion test shows this needs the Negative-Binomial correction rather than being false outright.
- The Bayesian trend model’s priors are weakly informative and were checked for sensitivity (Table 6) — the posterior does not depend meaningfully on the specific prior width chosen.
- The risk matrix’s “Almost certain” probability reading assumes the relevant question is *will at least one PUDO complaint occur somewhere in the fleet this quarter* (it will), not the odds for any single trip.
- Complaint counts are assumed complete and consistently defined within each reporting era; the 2024 aggregate table and the 2025+ ride-level microdata are assumed to measure the same underlying thing.

6.3 Limitations

- Complaints are self-reported by the carrier to its regulator; the rate measures *reported* problems, not necessarily underlying PUDO behavior, and reporting propensity could itself change over time.
- Waymo expanded to new cities during the window; a rate change may partly reflect new-market composition rather than a change in PUDO behavior.
- The 2024 and 2025+ numerators come from different source-file schemas, so a level shift at that boundary cannot be fully distinguished from a real trend.
- Redaction removes all within-quarter timing and location detail, and the travel-lane PUDO signal is unavailable in every file.
- Single carrier, single state, eight quarters — not yet a cross-carrier or multi-state comparison.

7 Conclusion and Future Work

Over 2024 Q3 – 2026 Q2, Waymo’s exposure-adjusted PUDO complaint rate fell sharply during the 2024 launch ramp and has been roughly flat since — not the rising trend a naive raw-count reading would suggest. On a probability $\times$ impact risk matrix the hazard currently sits in the **Medium**, monitor-and-mitigate band: frequent enough to expect every quarter, but low-consequence, with no severe or fatal PUDO events observed. These results are still provisional (see Limitations); as more CPUC quarters are published the same pipeline can extend the trend directly, and applying it to other CPUC-reporting AV carriers would turn a single-carrier trend into a cross-carrier comparison.

References

- [1] California Public Utilities Commission, “AV program quarterly reporting.” 2026. Available: <https://www.cpuc.ca.gov/regulatory-services/licensing/transportation-licensing-and-analysis-branch/autonomous-vehicle-programs/quarterly-reporting>